

Exp3code 2

```
clc,clear,close;
n1 = input("Enter RI of core:");
n2 = input("Enter RI of cladding:");
a = input("Enter radius of core:");
b = input("Enter radius of cladding:");
r = 0:1e-7:b;
L = length(r);
Delta = (n1^2 - n2^2)/(2*n1^2);
alpha = [1 2 3 15];
for j = 1:4
    for i = 1:L
        if r(i) <= a
            n(j,i) = n1 * sqrt(1 - 2*Delta*(r(i)/a)^alpha(j));
        else
            n(j,i) = n2;
        endif
    endfor
endfor
plot(r,n(1,:),r,n(2,:),r,n(3,:),r,n(4,:));
legend("alpha = 1", "alpha = 2","alpha = 3","alpha = 15");
axis([0 b n2-0.01 n1+0.01]);
xlabel("radial distance");
ylabel("Refractive Index");
title("RI vs radial distance for GI fiber");
```

Exp3code 1

```
clc,clear,close;
n1 = input("Enter RI of core:");
n2 = input("Enter RI of cladding:");
a = input("Enter radius of core:");
b = input("Enter radius of cladding:");
r = 0:1e-7:b;
L = length(r);
for i = 1:1:L
    if r(i) < a
        n(i) = n1;
    else
        n(i) = n2;
    endif
endfor
plot(r,n);
axis([0 b n2-0.01 n1+0.01]);
xlabel("radial distance");
ylabel("Refractive Index");
title("RI vs radial distance for SI fiber");
```

Exp6 code 2

Program 2 :

```
clc;
clear;
% Given values (in micro watts)
P1 = 60; % Input power
P2 = 0.004; % Output at port 2
P3 = 26.0; % Output at port 3
P4 = 27.5; % Output at port 4
% Excess loss
excess_loss = 10 * log10(P1 / (P3 + P4));
% Insertion loss (port 1 to port 4)
insertion_loss = 10 * log10(P1 / P4);
% Crosstalk
crosstalk = 10 * log10(P2 / P1);
% Split ratio
split_ratio = (P3 / (P3 + P4)) * 100;
% Display results
fprintf("Excess Loss = %.4f dB\n", excess_loss);
fprintf("Insertion Loss = %.4f dB\n", insertion_loss);
fprintf("Crosstalk = %.4f dB\n", crosstalk);
fprintf("Split Ratio = %.2f %%\n", split_ratio);
```

Exp6 code 1

Program 1 :

```
Clc,clear,close;
```

```
P1 = 100e6;
```

```
Lambda = 1000e-9;
```

```
N = 1.65;
```

```
K = 2pi/Lambda;
```

```
Z = 0:0.01e-9:500e-9;
```

```
P3 = P1*(cos(K.*Z)).^2;
```

```
P4 = P1(sin(K.*Z)).^2;
```

```
Plot(Z,P3/P1,"k-","z,P4/P1","k—");
```

```
Legend("P3/P1","P4/P1");
```

```
Xlabel("Coupler draw length in nm");
```

```
Ylabel("Power Ratios");
```

```
Title("Coupler Power Distribution");
```

Exp5 code 3

2ns:

```
clc,clear,close;
```

```
Pdc=300e-6;
```

```
f1=20e6;
```

```
f2=100e6;
```

```
f3=200e6;
```

```
w1=2*pi*f1;
```

```
w2=2*pi*f2;
```

```
w3=2*pi*f3;
```

```
tou=2e-9; %carrier recombination lifetime
```

```
Pew1=Pdc/sqrt(1+(w1*tou)^2);
```

```
Pew2=Pdc/sqrt(1+(w1*tou)^2);
```

```
Pew3=Pdc/sqrt(1+(w1*tou)^2);
```

```
disp("Output power(in uw) at 20MHz");
```

```
disp(Pew1*1e6)
```

```
disp("Output power(in uw) at 100MHz");
```

```
disp(Pew2*1e6)
```

```
disp("Output power(in uw) at 200MHz");
```

```
disp(Pew3*1e6)
```

```
BWopt1=sqrt(3)/(2*pi*tou);
```

```
disp("Optical BW in MHz")
```

```
disp(BWopt1*1e-6);
```

```
BWelect1=BWopt1/sqrt(2);
```

```
disp("electrical BW in MHz");
```

```
disp(BWelect1*1e-6);
```

Exp5 code 2

5ns:

```
clc,clear,close;
```

```
Pdc=300e-6;
```

```
f1=20e6;
```

```
f2=100e6;
```

```
f3=200e6;
```

```
w1=2*pi*f1;
```

```
w2=2*pi*f2;
```

```
w3=2*pi*f3;
```

```
tou=5e-9; %carrier recombination lifetime
```

```
Pew1=Pdc/sqrt(1+(w1*tou)^2);
```

```
Pew2=Pdc/sqrt(1+(w1*tou)^2);
```

```
Pew3=Pdc/sqrt(1+(w1*tou)^2);
```

```
Disp("Output power(in uw) at 20MHz");
```

```
Disp(Pew11e6)
```

```
Disp("Output power(in uw) at 100MHz");
```

```
Disp(Pew21e6)
```

```
Disp("Output power(in uw) at 200MHz");
```

```
Disp(Pew31e6)
```

```
BWopt1=sqrt(3)/(2*pi*tou);
```

```
Disp("Optical BW in MHz")
```

```
Disp(BWopt11e-6);
```

```
BWelect1=BWopt1/sqrt(2);
```

```
Disp("electrical BW in MHz");
```

```
Disp(BWelect11e-6);
```

Exp5 code 1

```
clc, close, clear;

Pdc = 300e-6; %Dc Power

f=1e6:1e3:300e6; %Frequency Range

w=2*pi*f;

tou1=5e-9;

Pew1=Pdc./sqrt(1+(w*tou1).^2);

PRatio1=Pew1./Pdc;

tou2=2e-9; %Carrier recombination lifetime

Pew2=Pdc./sqrt(1+(w*tou2).^2);

PRatio2=Pew2./Pdc;

plot(log10(f),PRatio1,"k",log10(f),PRatio2,"r" );

xlabel("Log of Frequency")

ylabel("Power Ratio");

title("Power Ratio Vs Frequency");

legend("tou1=5ns","tou2=2ns");
```

Exp4

```
clc,close,clear;
Pi = input(" Enter input power : ");
alpha = [0.2 ,0.5 ,2]; %distance in km
L = 0:0.01:20; %attenuation in dB/km
Po1 = Pi./(10.^(alpha(1)*L/10));
Po2 = Pi./(10.^(alpha(2)*L/10));
Po3 = Pi./(10.^(alpha(3)*L/10));
plot(L, Po1,"k:", L, Po2,"k-", L, Po3,"k--");
legend("alpha = 0.2 dB/km","alpha = 0.5 dB/km","alpha = 2 dB/km");
xlabel("Distance in km");
ylabel("O/P power in mW");
title("O/P power v/s Distance");
Ps = 2e-6; %receiver sensitivity
Pin = 1.5e-3; %input power
L1 = 10/alpha(1)*log10(Pin/Ps);
disp("Max. Repeatless distance at alpha = 0.2db/km");
disp(L1);
L2 = 10/alpha(2)*log10(Pin/Ps);
disp("Repeatless distance at alpha = 0.5db/km");
disp(L2);
L3 = 10/alpha(3)*log10(Pin/Ps);
disp("Repeatless distance at alpha = 2db/km");
disp(L3);
```

Exp2 code 3

```
Clc,clear,close;
```

```
A = input("enter core radius:");
```

```
Lambda = 800:1:1600; % in nm
```

```
NA = [0.1 0.15 0.2];
```

```
V1 = (2*pi./(lambda*1e-9))*a*NA(1);
```

```
V2 = (2*pi./(lambda*1e-9))*a*NA(2);
```

```
V3 = (2*pi./(lambda*1e-9))*a*NA(3);
```

```
Plot(lambda,V1,"k:",lambda,V2,"k-",lambda,V3,"k—");
```

```
Legend("NA = 0.1","NA = 0.15", "NA =0.2");
```

```
Xlabel("wavelength in nm");
```

```
Ylabel("V number");
```

```
Title("V number as function of wavelength");
```

Exp2 code 2

Clc,clear,close;

D = input("enter core diameter:");

N1 = input("enter RI of core:");

Delta = input("enter RRID:");

A = d/2;

NA = n1sqrt(2delta);

Vc = 2.405; % for single mode SI fiber

Lambda = 2pi/Vca*NA;

Disp("shortest wavelength");

Disp(lambda);

Exp2 code 1

```
Clc,clear,close;

Lambda1 = input("enter first wavelength:");

Lambda2 = input("enter second wavelength:");

Lambda3 = input("enter third wavelength:");

A = input("enter core radius:");

N1 = input("enter RI of core:");

N2 = input("enter RI of cladding:");

NA = sqrt(n1^2 - n2^2);

Disp("Numerical Aperture");

Disp(NA);

V1 = (2pi/Lambda1)aNA;

Disp("V number at 850nm");

Disp(V1);

Ms1 = V1^2/2;

Disp("No. of modes at 850nm");

Disp(Ms1);

V2 = (2pi/Lambda2)aNA;

Disp("V number at 1300nm");

Disp(V1);

Ms2 = V1^2/2;

Disp("No. of modes at 1300nm");

Disp(Ms2);

V3 = (2*pi/Lambda3)aNA;

Disp("V number at 1550nm");

Disp(V1);

Ms3 = V1^2/2;

Disp("No. of modes at 1550nm");

Disp(Ms3);
```

Exp1 code 3

```
Clc,clear,close;
```

```
thetaA=input("Enter given acceptance angle:");
```

```
delta=input("Enter refractive index difference:");
```

```
NA = sind(thetaA);
```

```
Disp("Numerical Aperture")
```

```
Disp(NA);
```

```
N1= NA / sqrt(2delta);
```

```
Disp("RI of core");
```

```
Disp(n1);
```

```
N2=n1(1-delta);
```

```
Disp("RI of cladding");
```

```
Disp(n2);
```

```
phiC=asind(n2/n1);
```

```
display("Critical angle in degree.")
```

```
disp(phiC);
```

Exp1 code 2

```
Clc,clear,close;
```

```
V1=input("velocity in core:");
```

```
Phic=input("Enter critical angle in degree:");
```

```
Phic=phicpi/180;
```

```
C=3e8;
```

```
N1=c/v1;
```

```
Disp("RI of core")
```

```
Disp(n1);
```

```
N2 = n1sin(phic);
```

```
Disp("RI of cladding");
```

```
Disp(n2);
```

```
NA = sqrt(n1^2-n2^2);
```

```
Disp("Numerical Aperture");
```

```
Disp(NA);
```

```
N0 = 1;
```

```
thetaA = asin(NA/n0);
```

```
disp("acceptance angle in degree");
```

```
disp(thetaA*180/pi);
```

Exp1 code 1

```
Clc,clear,close;

N1=input("Enter RI core:")

N2=input("Enter RI cladding:")

NA = sqrt(n1^2-n2^2)

Disp("Numerical Aperture")

Disp(NA);

phiC=asin(n2/n1);

display("Critical angle in degree.")

disp(phiC180/pi);

n0 = 1;

thetaA = asin(NA/n0);

disp("acceptance angle in degree");

disp(thetaA180/pi);
```